ModelIt!

Teacher's Guide

Can CRISPR Cure Genetic Diseases?

Modeling Gene Editing Precision, Off-Target Effects, and Ethical Boundaries

Quick Reference

Grade Level	Grade 11L3
NGSS Standard	HS-LS3-1, HS-LS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS3-1, HS-LS3-2

Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring. Make and defend a claim based on evidence that inheritable genetic variations may result from new genetic combinations through meiosis, viable errors occurring during replication, or mutations caused by environmental factors.

Three-Dimensional Learning

Science Practice	**Asking Questions and Defining Problems** — Students ask questions about how CRISPR precision, off-target rates, and repair pathways interact to determine whether gene editing therapy succeeds or causes harm.
Disciplinary Core Idea	**LS3.A Inheritance of Traits / LS3.B Variation of Traits** — Students model how DNA mutations cause genetic diseases and how targeted gene editing can correct these mutations, while also understanding how editing introduces new genetic variation.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chain from guide RNA design through Cas9 cutting, repair pathway selection, and ultimately therapeutic or harmful outcomes, identifying how each variable influences the final result.

CAST Testing Alignment

MULTIPLE CHOICE:

A CRISPR therapy achieves 95% on-target editing efficiency but has a 0.1% off-target rate. Given that the human genome has 3.2 billion base pairs, approximately how many unintended edits could occur per treated cell, and why does this matter?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why curing sickle cell disease with CRISPR in adult blood stem cells is considered ethically acceptable by most scientists, while using the same technology to edit embryos to prevent sickle cell disease in future generations remains deeply controversial. Reference at least two model variables in your response.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

CRISPR-Cas9 is the most transformative biological technology since the discovery of DNA's structure in 1953. Derived from an ancient bacterial immune system, it allows scientists to edit any gene in any organism with unprecedented precision. In 2023, the FDA approved the first CRISPR-based therapy (Casgevy) for sickle cell disease, marking the beginning of the gene editing era in medicine. But the power to rewrite the code of life raises questions that science alone cannot answer.

How CRISPR-Cas9 Works

The CRISPR system has two key components: a guide RNA that matches the target DNA sequence, and the Cas9 enzyme that acts as molecular scissors. The guide RNA directs Cas9 to the target location by base-pairing with the complementary DNA strand. Once bound, Cas9 makes a double-strand break in the DNA. The cell then repairs this break using one of two pathways: non-homologous end joining (NHEJ), which is fast but error-prone and often introduces small insertions or deletions, or homology-directed repair (HDR), which uses a provided DNA template to make precise corrections. Scientists prefer HDR for therapeutic applications, but cells default to NHEJ about 80% of the time.

The Off-Target Problem

The human genome contains 3.2 billion base pairs, and many 20-nucleotide sequences appear multiple times with slight variations. A guide RNA designed to match one specific location may also bind to sequences that differ by just one or two bases. When Cas9 cuts at these off-target sites, it can disrupt essential genes, activate cancer-causing oncogenes, or create chromosomal rearrangements. Modern high-fidelity Cas9 variants (like eSpCas9 and HiFi Cas9) reduce but do not eliminate off-target activity. Whole-genome sequencing after editing reveals that even optimized systems produce detectable off-target mutations.

Somatic vs. Germline: The Ethical Divide

Somatic gene editing modifies cells in a living patient — blood cells, muscle cells, neurons — and those changes die with the patient. This is ethically similar to any medical treatment: the patient consents, bears the risks, and receives the benefits. Germline editing modifies embryos, sperm, or eggs, meaning the changes become heritable and pass to all future generations. In 2018, Chinese scientist He Jiankui created the first gene-edited babies, modifying the CCR5 gene to confer HIV resistance. The global scientific community condemned this as premature, reckless, and unethical — he was sentenced to three years in prison. The ethical debate centers on consent: future generations cannot consent to changes that permanently alter the human genome.

Current Clinical Applications

As of 2025, CRISPR therapies are approved or in late-stage clinical trials for sickle cell disease, beta-thalassemia, certain cancers (via CAR-T cell engineering), hereditary blindness (Leber congenital amaurosis), hereditary angioedema, and transthyretin amyloidosis. The sickle cell therapy works by editing blood stem cells outside the body, reactivating fetal hemoglobin production that compensates for the defective adult hemoglobin gene. Results have been remarkable: patients who suffered debilitating pain crises are now symptom-free years after treatment. However, the therapy costs over $2 million per patient, raising serious questions about equitable access.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven key components of the CRISPR gene editing system: guide RNA specificity, Cas9 cutting efficiency, off-target edit rate, repair pathway selection, delivery efficiency, therapeutic benefit, and unintended mutations.

E **Establish Relationships**

Students map the causal relationships showing how guide RNA specificity and delivery efficiency drive both therapeutic benefit and off-target risk, revealing the fundamental precision-versus-coverage trade-off in gene editing.

V **Visualize & Model**

Students build a computational model showing the seven-component gene editing system with competing pathways leading to therapeutic success or unintended harm.

E **Evaluate Outcomes**

Students run high-precision sickle cell, low-specificity broad edit, and germline versus somatic comparison scenarios to identify the parameter ranges where therapeutic benefit outweighs risk.

R **Revise & Extend**

Students add epigenetic disruption, immune response, or mosaicism rate to model more realistic biological complexity in CRISPR therapy outcomes.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In 2023, a 12-year-old named Victoria Gray became the first person cured of sickle cell disease by CRISPR gene editing. She had suffered agonizing pain crises her entire life. Now she is pain-free. But the same technology that saved her could also be used to edit human embryos — changing the DNA of every future generation. Where do we draw the line?"*

Watch For

Open with Victoria Gray's story. Show before and after — a life of pain crises versus a cure. Let the emotional weight settle before introducing the technology.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the most powerful biological tool humans have ever created. You will discover why it works, where it fails, and why the hardest questions about CRISPR are not scientific but ethical."*

Watch For

Have students read objectives. Pre-teach 'guide RNA' and 'off-target effects.' Emphasize that precision is the central engineering challenge.

Slide 3: Big Question

3 min

Say: *"Notice this question has two parts: CAN we and SHOULD we. The science answers the first. Only society can answer the second. Today your model will show you exactly why that distinction matters."*

Watch For

Quick-write: Students write what they already know about CRISPR and one concern they have about gene editing. Share out and catalog concerns on the board.

Slide 4: LEVER Framework

2 min

Say: *"We are building a model of how molecular precision at the scale of individual DNA bases determines whether a patient is cured or harmed."*

Watch For

Preview each LEVER step. Emphasize that this model connects molecular biology to medical outcomes to ethical decisions.

Slide 5: Activity 1: Components

8 min

Say: *"Seven components control whether gene editing saves a life or causes new damage. Two of them — guide RNA specificity and delivery efficiency — are the variables scientists can directly engineer. Everything else is a consequence."*

Watch For

Guide students through components. Discuss why guide RNA specificity and delivery are external — they are the design choices. Cutting efficiency, repair pathway, and outcomes are biological responses.

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the cruel trade-off: the same guide RNA that finds your target also searches the entire genome for similar sequences. More delivery means more editing — both on-target AND off-target."*

Watch For

Help students map the dual-pathway structure. Use green arrows for therapeutic pathways and red for harmful ones. Emphasize that both flow from the same input variables.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us test: can you find parameter settings where the therapy cures the disease without creating dangerous new mutations? What happens when you push for maximum coverage?"*

Watch For

Students predict outcomes before each scenario. Compare predictions with model outputs. Special attention to the germline scenario — same numbers, completely different ethical weight.

Slide 8: Discoveries

5 min

Say: *"CRISPR is not a magic eraser — it is a precision tool operating in a genome with 3.2 billion characters where many sequences look alike. The question is not IF off-target edits happen but WHETHER the therapeutic benefit outweighs the risk."*

Watch For

Lead discussion connecting model findings to real clinical outcomes. Victoria Gray is cured — but the therapy costs two million dollars and required destroying and rebuilding her blood system. Is that sustainable?

Slide 9: STEM Challenge

12 min

Say: *"A biotech company needs your clinical trial design. Fatal disease, promising therapy, known risks. How do you protect patients while advancing a cure that could save millions?"*

Watch For

Groups design clinical trial protocols addressing target selection, safety monitoring, patient selection, and ethical review. Present with evidence from model simulations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Guide RNA Specificity, Delivery Efficiency
- **INTERNAL:** Cas9 Cutting Efficiency, Off-Target Edit Rate, Repair Pathway Selection, Therapeutic Benefit, Unintended Mutations

Relationships

- **Guide RNA Specificity to Off-Target Edit Rate:** NEGATIVE (-) — Higher guide RNA specificity means the guide RNA binds ONLY to the intended target and not to similar sequences — reducing off-target cuts. Lower specificity allows binding at multiple genomic locations, increasing unintended edits.
- **Delivery Efficiency to Cas9 Cutting Efficiency:** POSITIVE (+) — Higher delivery efficiency means more cells receive the CRISPR components, increasing the percentage of target cells where Cas9 successfully makes double-strand breaks at the intended site.
- **Off-Target Edit Rate to Unintended Mutations:** POSITIVE (+) — Higher off-target editing means Cas9 is cutting DNA at more unintended locations, directly increasing the number of novel mutations that could disrupt genes, activate oncogenes, or cause chromosomal rearrangements.
- **Cas9 Cutting Efficiency to Therapeutic Benefit:** POSITIVE (+) — Higher cutting efficiency at the target site means more cells have the disease-causing mutation corrected, directly increasing the measurable improvement in the patient's condition.

Key Discoveries

- CRISPR precision is extraordinary by historical standards but still imperfect — even the best guide RNAs produce detectable off-target edits, and the 3.2 billion base-pair genome provides countless near-match sites
- The cell's DNA repair choice after Cas9 cutting is largely outside the scientist's control — error-prone non-homologous end joining is more common than precise homology-directed repair, meaning many edits introduce random mutations rather than the intended correction
- Delivery efficiency creates a dose-response dilemma — increasing CRISPR components to edit more cells also increases off-target events proportionally, and not all tissues are equally accessible
- The ethical boundary between curing disease and enhancing traits is scientifically indistinguishable — the same CRISPR technology that corrects sickle cell disease could theoretically modify intelligence, athleticism, or appearance genes

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"CRISPR edits genes with perfect precision like a word processor"	CRISPR is remarkably precise compared to previous genetic engineering tools, but it is not perfect. The Cas9 enzyme can cut at off-target sites that partially match the guide RNA, the cell's DNA repair after cutting is not fully controllable (NHEJ is error-prone and more common than HDR), and large deletions or chromosomal rearrangements occasionally occur at cut sites. Current best-case on-target efficiency is about 90-95% with detectable off-target activity at multiple genomic sites.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Gene editing can only be used for curing diseases"	The same CRISPR technology that corrects disease-causing mutations can in principle modify any gene — including those affecting height, intelligence, athletic ability, skin color, or any other trait. The technology does not distinguish between therapeutic and enhancement applications. This is why the scientific community draws a sharp ethical line between somatic editing for medical treatment and germline editing that could enable designer babies.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If CRISPR can cure one genetic disease it can cure them all"	Different genetic diseases present vastly different CRISPR challenges. Single-gene disorders with well-characterized mutations in accessible tissues (like sickle cell in blood cells) are the easiest targets. Diseases involving multiple genes, genes in hard-to-reach tissues (brain, heart), or conditions where the gene function is complex and context-dependent are far harder. Some genetic conditions would require editing millions of cells in solid organs simultaneously, which is currently impossible. The model shows that delivery efficiency and tissue accessibility are major limiting variables.